



INTERNATIONAL INSTITUTE OF
INFORMATION TECHNOLOGY

HYDERABAD

OUTREACH DIVISION

Web Scrapping



Access data from Websites

- **Web scraping** is a technique for gathering data or information from any public website.
- We can extract any data which we can see while browsing the web.
- We can save the data to a local file in your computer or to a database in table (spreadsheet) format using Pandas library.

Web Scraping

```
< HTML>  
    <HEAD>  
    .....  
    </HEAD>  
    <BODY>  
    .....  
    .....  
    </BODY>  
</HTML>
```

Website pages,
Unstructured Data



Structured /
Semi-structured
Data

Web Scraping Workflow

Request and Load Webpage

- Send Request
- Load the webpage (Requests, urllib)

Parsing

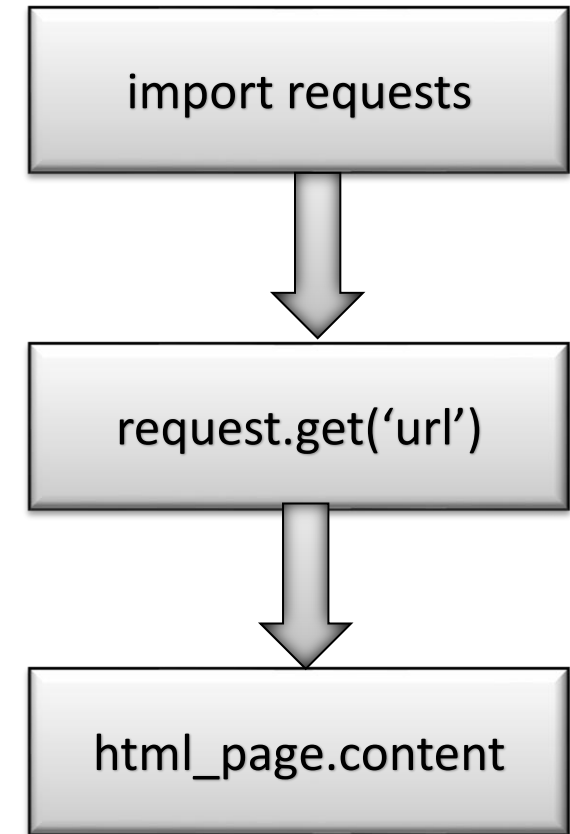
- Parse the content for desired data (Beautiful Soup)

Storing Data

- Store the data the way you want for later usage (Regular expression)

Performing HTTP Request in Python

- Import requests module
- Download the html_page from the specified path using requests module
- Get the content from the downloaded html_page

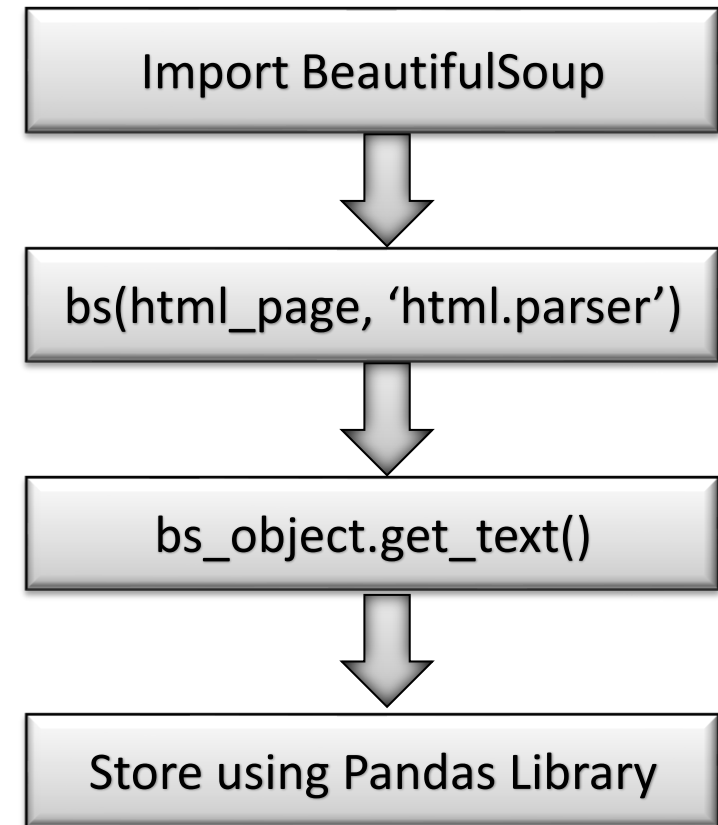


Parsing the data

- HTML is nested, cannot extract data simply through string processing.
- Need parser to create a tree structure of the HTML data.
- Use **Beautiful Soup** for parsing the HTML data.
- We can extract any specific data like author name, title, tables, and description.

Beautiful Soup

- Import BeautifulSoup
- Parse the html_page
- Extract the text without any HTML Tags
- Store the data in a required format



Experiment

➤ Demo Web Scraping

Objective: Extract data from web page using BeautifulSoup

Natural Language Toolkit (NLTK)



NLTK Package

- One of the most powerful NLP libraries which contains packages to make machines understand human language
- Tokenization helps in interpreting the meaning of the text by splitting string, text into a list of tokens.
- **"Punkt"** module for splitting the paragraphs into sentences, sentences into words.
- Wordnet is a tool integrated into NLTK that contain listings of word relations.
- ImageNet is a largescale ontology of images built upon the backbone of the WordNet structure. ImageNet constitutes ~ 10% of the "WordNet.synsets"

NLTK Basic Functionalities

- Sentence Tokenization
- Word Tokenization
- Parts of Speech (POS) tagging
- Stemming and Lemmatization
- Stop words removal

Text Analytics

- **Sentence Tokenization** means breaking down a paragraph into sentences.
- **Word Tokenization** means breaking down a paragraph into words.
- **Parts of Speech (POS) Tagging** looks for relationship within the sentence and assigns label for each word with its appropriate parts of speech.

Pre-Processing Text

- **Stemming** reduces the word to the root word
 - was -> wa; studies -> studi; studying -> study
- **Lemmatization** is similar to stemming but is related to grammatical context
 - was -> be; studies -> study; studying -> study
- **Stop words:** Noise in the text. Text may contain stop words such as is, am, at, of, are this, a, the, an.
- Remove these stop words from the list of words

Experiment

➤ Demo NLTK

Objective: Text pre-processing using NLTK

Thanks!!

Questions?